

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)

ORGANISATION OF ISLAMIC COOPERATION (OIC)

Department of Computer Science and Engineering (CSE)

MID SEMESTER EXAMINATION

WINTER SEMESTER, 2023-2024

DURATION: 1 HOUR 30 MINUTES

FULL MARKS: 75

Math 4341: Linear Algebra

Programmable calculators are not allowed. Do not write anything on the question paper.

Answer all 3 (three) questions. Figures in the right margin indicate full marks of questions with corresponding COs and POs in parentheses.

1. a) Considering the following system of linear equations, answer the questions that comes after: 1 + 1 + 1 + 3
(CO1)
(PO1)
- $$x + ky = 1$$
- $$kx + y = 1$$
- For what value of k , there exists no solution?
 - For what value of k , there exists infinitely many solutions?
 - For what value of k , there exists exactly one solution?
 - Determine the value of x and y in the case there is exactly one solution.
- b) A linear transformation function takes a shape in a 2D space (as shown in Figure 1) as input, performing a rotation of 90 degrees clockwise and then shrinking it down to half of its original size. 4 + 5
(CO1)
(PO1)

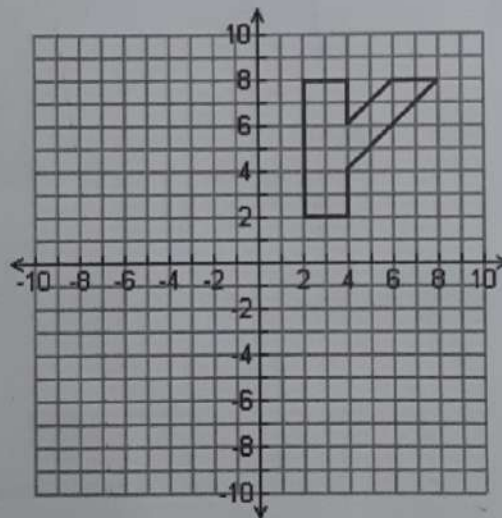


Figure 1: A shape in 2D space

- Given the original basis vectors, $\hat{i} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and $\hat{j} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$, Determine the transformed basis vectors after the specified transformation.
 - Roughly sketch the transformed shape which goes through the aforementioned transformation.
- c) Given the matrix A , find its inverse A^{-1} using the Gauss-Jordan method, starting with the augmented matrix $[A \mid I_3]$, where I_3 is the identity matrix of order 3×3 . 10
(CO1)
(PO1)

$$A = \begin{bmatrix} 1 & 2 & -1 \\ -2 & 0 & 1 \\ 1 & -3 & 1 \end{bmatrix}$$

2. a) You need to solve the following system of linear equations using Gaussian Elimination by answering the questions that come after:

$$\begin{cases} a + 2b - c + 5d = 22 \\ 3a + 6b - 3c - 2d = -2 \\ 7a - 5b + 3c - d = 2 \\ 5a + 10b + 2c - 7d = 3 \end{cases}$$

- Write the given system of linear equations in $Ax = b$ or matrix form.
 - Write the augmented matrix $[A|b]$.
 - Transform $[A|b]$ into $[U|c]$ where U is an upper triangular matrix. Use a series of elimination steps and, if necessary, permutations, using elimination matrices E_{ij} and permutation matrices P_{ij} . Note that the elimination matrix E_{ij} eliminates the value at position (i, j) of the matrix using pivot at position (i, i) . And permutation matrix P_{ij} exchanges row i with row j .
 - Determine the values of a, b, c and d using back substitution.
- b) Tony Stank has developed a new high-tech suit equipped with four engines. Each engine propels the suit in a specific direction, allowing Mr. Stank to fly in any direction by adjusting the power output of each engine. The directions of the four engines are represented by the following vectors:

$$a_1 = \begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix} \quad a_2 = \begin{bmatrix} -1 \\ 1 \\ 10 \end{bmatrix} \quad a_3 = \begin{bmatrix} 0 \\ 5 \\ -2 \end{bmatrix} \quad a_4 = \begin{bmatrix} -7 \\ 4 \\ 1 \end{bmatrix}$$

Due to unexpected fuel shortages, Mr. Stank must reduce the suit's fuel consumption. Determine the maximum number of engines Mr. Stank can shut down while still maintaining the ability to move freely in 3D space, and identify which engine(s) he can turn off without losing this capability of moving freely in 3D space.

3. a) Let P be the vector space formed by the linear combination of the vectors $\begin{bmatrix} -2 \\ 1 \\ 5 \end{bmatrix}$ and $\begin{bmatrix} 3 \\ -2 \\ -2 \end{bmatrix}$.

Q is the vector space formed by the vectors $c \begin{bmatrix} 4 \\ -3 \\ 1 \end{bmatrix}$, where c is any scalar.

Comment on whether $(P \cup Q)$ and $(P \cap Q)$ will be a vector space with proper justification.

- b) Consider the following two matrices in Reduced Row Echelon Form (RREF).

$$\text{RREF}(B) = \begin{bmatrix} 1 & 5 & 0 & 2 & -2 & -14 & 0 \\ 0 & 0 & 1 & -3 & -6 & 2 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \quad \text{RREF}(C) = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Answer the following questions for each of the given matrices:

- Identify the pivot columns and the free columns.
- Determine the rank of the given matrix.
- Determine the bases of its null space.
- What will be the nature of null space obtained in Question 3.b)iii?

c) You have been provided with a matrix,

$$M = \begin{bmatrix} 1 & -2 & -3 & 4 & -5 \\ 0 & 2 & 204 & 117 & 238 \\ 0 & 0 & 1 & 201 & 240 \\ 0 & 0 & 0 & -2 & 212 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

4
(CO1)
(PO1)

Let Z be another a 5×5 matrix representing a volume in a 5D space. If matrix Z is multiplied by matrix M , how will the volume and the shape it occupies in the 5D space change?